

Examen Parcial 2

1) $V_{pt} = 25 \text{ m/s}$ $a = -0.7 \text{ m/s}^2$ $X_0 = 0 \text{ m}$

$$X = 0 + V_0 t + \frac{1}{2} a t^2 \quad \frac{1}{2}(-0.7) = -0.35 \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{Parajeros}$$

$$X_{pt} = 25 \text{ m/s} t - 0.35 \text{ m/s}^2 t^2$$

$V_{ft} = 15 \text{ m/s}$ $a = 0 \text{ m/s}^2$ $X_0 = 200 \text{ m}$ $\left. \begin{array}{l} \\ \end{array} \right\} \text{Carga}$

$$X = X_0 + V_0 t + \frac{1}{2} a t^2$$

$$X_{ft} = 200 \text{ m} + 15 \text{ m/s} t$$

$$X_{pt} = X_{ft}$$

$$25 \text{ m/s} t - 0.35 \text{ m/s}^2 t^2 = 200 \text{ m} + 15 \text{ m/s} t$$

$$0.35 \text{ m/s}^2 t^2 - 10 \text{ m/s} t + 200 \text{ m} = 0 \rightarrow \text{Cuadrática}$$

$$t_1 = 77.5 \text{ s}$$

$$t_2 = -22.5 \text{ s}$$

$$X_{FT} = 200 \text{ m} + 15 \text{ m/s} (22.5) = 537.5 \text{ m}$$

La colisión ocurre en la posición 537.5 m

2) $V_f^2 - V_i^2 = 2ad$

$$V_i = 22 \text{ m/s}$$

$$V_f = 0$$

$$d = 120 \text{ m}$$

$$0 - (22 \text{ m/s})^2 = 2(120 \text{ m})a$$

$$-484 \text{ m/s}^2 = 240 \text{ m} a$$

$$a = \frac{-484 \text{ m/s}^2}{240 \text{ m}} = -2.0166 \approx -2.02 \text{ m/s}^2$$

$$3) \quad V_1 = 47 \text{ km/h} (-\cos 60^\circ i + \sin 60^\circ j)$$

$$3 \text{ horas después: } V_2 = 25 \text{ km/h} (j)$$

$$t = 3h + t_2 \quad t_2 = 4.5h - 3h$$

$$t_2 = 1.5h$$

$$d = Vt \quad d = 47 \text{ km/h} (3h) (-\cos 60^\circ i + \sin 60^\circ j) + 25 \text{ km/h} (1.5h j)$$

$$d = 123 \text{ km} (-\cos 60^\circ i + \sin 60^\circ j) + 37.5 \text{ km} j$$

$$d = -61.5 i + 106.52 j + 37.5 j$$

$$d = -61.5 i + 144.02 j \text{ km} \rightarrow \text{rect. desplazamiento}$$

$$d = \sqrt{(-61.5)^2 + (144.02)^2}$$

$$d = 156.60 \text{ m}$$

4)

$$3 + 4 = 7$$

$$\frac{7}{10} = 0.70 \text{ m/s}$$

$$5) \quad V_f - V_0 = V_0 t + \frac{1}{2} a t^2$$

$$40 \text{ m} = \frac{1}{2} (2.70 \text{ m/s}^2) t^2$$

$$40 \text{ m} = (27/20) t^2$$

$$t^2 = \frac{800}{27}$$

$$27$$

$$t = \sqrt{\frac{800}{27}} \approx 6.175$$

} Dist. que adelanta el camión

$$x_f = x_f$$

$$V_0 + V_0 t + \frac{1}{2} a t^2 = x_0 + V_0 t + \frac{1}{2} a t^2$$

$$\frac{1}{2} (3.4 \text{ m/s}) (6.77)^2 = V_0 + \frac{1}{2} (2.70 \text{ m/s}) (6.77)^2$$

$$V_0 = 24.7447 \approx \underline{24.74 \text{ m}}$$